

Internship proposal

Evaluating Prior-Trained Causal Effect Estimation

Context: Estimating causal effects from observational data generally requires assumptions about the underlying causal structure. Classical approaches first establish whether a causal effect is identifiable from a causal graph and then estimate the resulting functional, for instance using adjustment or the g-formula Pearl [2009], Hernan and Robins [2023].

Prior-Data Fitted Networks (PFNs) provide a different paradigm, in which a transformer is pre-trained on data sampled from a prior over data-generating processes and subsequently performs inference in context Müller et al. [2022].

Recently, Do-PFN proposed a different approach: a transformer is pre-trained on data generated from synthetic structural causal models and subsequently predicts interventional distributions from observational data without requiring the causal graph at inference time.

This approach is promising, but it raises important questions about the role of the synthetic pre-training distribution. In particular, its performance may depend on how closely the target data-generating process matches the structural and functional assumptions encoded in the pre-training prior. Moreover, existing experiments provide only limited comparison with classical identification-based estimators that explicitly exploit knowledge of the true causal graph Robertson et al. [2025].

Proposal: The main objective of this internship is to empirically investigate when prior-trained causal effect estimators such as Do-PFN are competitive with, or provide an advantage over, classical causal inference methods.

We will first construct controlled simulation settings in which the true causal graph is known and the causal effect is identifiable. Do-PFN will be compared with classical graph-based estimators that explicitly exploit this knowledge, including adjustment-based estimators and g-computation.

The study will then progressively depart from the assumptions encoded in the pre-training distribution in order to investigate the robustness of Do-PFN, considering:

- different functional forms and noise distributions;
- treatment–outcome pairs with weak, null, or indirect causal effects;
- observed and unobserved confounding;
- identifiable and non-identifiable settings;
- different graph and sample sizes.

In non-identifiable settings, particular attention will be given to understanding the role of the pre-training prior and the uncertainty produced by Do-PFN.

Finally, this analysis will be extended to time series by considering recent extensions of prior-trained causal inference to temporal settings Thumm and Chen [2026]. In particular, we will investigate causal effect estimation on time series generated from discrete-time structural causal models Assaad et al. [2024], Ferreira and Assaad [2024], Assaad [2025]. This will allow us to study whether the conclusions obtained in the static setting continue to hold when causal relationships involve temporal dependencies and lagged effects.

The internship aims to provide a clearer understanding of the trade-off between two approaches to causal effect estimation:

causal graph $\longrightarrow$ identification $\longrightarrow$ estimation,

and

prior over SCMs $\longrightarrow$ pre-training $\longrightarrow$ in-context causal inference.

The project should help determine when prior-trained causal models can complement classical identification-based methods and when explicit causal knowledge remains preferable.

Candidate: Highly motivated candidate with an M2 degree and strong background in probability, machine learning, and causal inference, along with a keen interest in epidemiology. Proficiency in programming is also required.

Location: The intern will work at Sorbonne Public health Institute (<https://iplesp.fr/>), located in Paris. They will be co-supervised by Charles Assaad, Pierre-Yves Boëlle, and Eugenio Valdano.

Dates: Starting date: To be discussed, early 2027, for a duration of 5-6 months.

Contact: To apply, please send a CV and a cover letter to Charles Assaad charles.assaad@inserm.fr

References

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